Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 281 (2026) 1463–1468

Procedia Computer Science

www.elsevier.com/locate/procedia

The 6th International Conference on Multi-modal Information Analytics (MMIA)

Consumer Behavior Data Mining and Analysis Using Machine Learning Algorithms

Hanwen Zhang[a]* , Yueyue Hou[b]

[a]*College of Business, Cheongju University, Cheongju, Republic of Korea*
[b]*Guangzhou Software Institute, Guangzhou, China*

Abstract

In the era of digital economy, the vast amount of consumer online behavior data provides unprecedented possibilities for accurate insight into market demand and prediction of individual behavior. This study aims to systematically explore and compare the effectiveness of different machine learning algorithms in consumer behavior data mining and analysis. Focusing on the core task of "prediction of customers' future purchase intention", the research selects four typical algorithms, including logical regression, support vector machine, random forest and XGboost, and constructs a complete analysis process from data preprocessing, feature engineering to model training evaluation on a real e-commerce data set. This paper systematically reviews the evolution from classical behavior theory to modern data mining technology. In terms of methodology, this paper describes the key steps of experimental conditions, data cleaning, feature construction (including RFM and extended features) and model implementation in detail. The experimental results are presented clearly through the comprehensive performance table, efficiency comparison table and feature importance table. The analysis shows that XGboost algorithm performs best in accuracy, F1 score, AUC and other key indicators, showing a strong ability to deal with complex nonlinear relationships; The Stochastic Forest achieves a good balance in stability and efficiency; However, logistic regression maintains the best explicability. This study not only verifies the superiority of ensemble learning in consumer behavior prediction, but also provides empirical basis and selection guidance for enterprises in the trade-off between accuracy, efficiency and interpretability.

Keywords: Machine learning, Consumer behavior analysis, Data mining, Predictive model, XGBoost, RFM model;

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000 .
E-mail address: zhw0314@gmail.com

1. Introduction

With the rapid development of e-commerce, mobile Internet and Internet of things technology, a series of consumer behaviors such as browsing, searching, clicking, buying and evaluating have been completely recorded on the digital platform, forming a huge, diverse and real-time big data resource. This marks a profound transformation of the research paradigm of consumer behavior from causal inference based on sampling survey to correlation mining and predictive analysis based on full data [1].

However, in the face of such complex data forms, the traditional statistical analysis tools (such as linear regression and analysis of variance) have become inadequate. As the core branch of artificial intelligence, machine learning automatically learns patterns and rules from data through algorithms, and shows significant advantages in dealing with high-dimensional, nonlinear and complex big data. In recent years, machine learning has been widely used in many business scenarios, such as customer churn prediction, marketing response modeling, credit scoring and so on, from classic logical regression and support vector machines to modern integrated learning algorithms (such as random forest and gradient lifting tree) [2]. In view of this, this study aims to empirically compare and deeply analyze four mainstream machine learning algorithms through a rigorous and reproducible data science process. We not only pay attention to the prediction accuracy of the model, but also incorporate engineering practice indicators such as training efficiency and feature importance into the comprehensive evaluation system, in order to provide researchers and practitioners in related fields with a clear and comprehensive algorithm performance map and selection decision reference.

2. Related Works

Data driven research on consumer behavior is a cross field integrating marketing, computer science and statistics. Its development closely follows the evolution of theoretical basis, data form and analysis technology.

In terms of analysis methods, early data mining technologies focused on discovering patterns from structured transaction data. The Apriori algorithm proposed by Lin J[3] and others is a milestone in association rule mining. It makes it possible to automatically discover the symbiotic relationship between "beer and diapers" from large-scale transaction data, and is widely used in shopping basket analysis and cross selling. The FP growth algorithm proposed by Ebrahimi P[4] and others has greatly improved the efficiency of association rule mining through novel data structure. For sequential patterns, the sequential pattern mining algorithm proposed by Mohan L[5] can analyze the purchase order of customers across time periods and provide a tool for predicting the next possible purchase. In the field of customer segmentation, the K-means clustering algorithm and its subsequent variants proposed by Chaubey G[6] and others have become one of the most commonly used unsupervised learning methods for market clustering based on customer attributes or behavior characteristics. These traditional methods are good at descriptive analysis and pattern discovery, but their ability is often limited in complex predictive modeling.

The rise of machine learning has brought a paradigm level breakthrough for consumer behavior analysis. In the field of supervised learning, classification and regression prediction have become the core applications. The support vector machine proposed by Li H[7] has been widely used in customer classification and text sentiment analysis because of its solid foundation based on statistical learning theory and the ability to handle nonlinear problems through kernel functions. The random forest algorithm proposed by Bhoyar S[8] and others has significantly improved the accuracy and robustness of the model by integrating multiple decision trees and introducing randomness. It is widely used in customer churn prediction and credit risk assessment. The concept of gradient elevator proposed by Abdul Aziz M[9] and Zvarikova K[10] and others, as well as the efficient implementation of XGboost, have achieved the performance of state of the art in many data science competitions and industrial scenes by iteratively optimizing the loss function and integrating weak learners, and become one of the preferred algorithms for processing table data.

In recent years, with the increase of data complexity, more advanced models have been introduced. For the temporal dependence of behavior sequence, the long-term and short-term memory network and its variants proposed by Ghorbantanhaei H[11] and others can effectively capture long-term dependence and be applied to the next recommendation and loss warning. Xu Z[12] and others successfully applied deep reinforcement learning in the field of go, which also inspired their application and exploration in sequential decision-making problems such as personalized marketing and dynamic pricing.

To sum up, the existing research has fully proved the great value of machine learning in consumer behavior analysis, and has developed from linear model and kernel method to today's deep learning and integrated learning. This study aims to fill this gap, through rigorous controlled experiments, quantitative evaluation of logistic regression SVM. The comprehensive performance of random forest and XGboost under the same data and evaluation system, and the root causes of their performance differences are discussed in depth.

3. Method

3.1. Data sources and experimental conditions

This study used the publicly available "Online Retail" dataset from the UCI machine learning repository. This dataset records all cross-border transactions of an online retail company from December 1, 2020 to December 9, 2021, including fields such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, Customer ID, and Country.

The hardware environment for the experiment is an Intel Core i7-12700H processor with 16GB DDR4 memory. The software environment is Python 3.9, with main libraries including Pandas 1.4.2, NumPy 1.22.3, Scikit learn 1.0.2, XGBoost 1.5.0, and visualization using Matplotlib and Seaborn.

3.2. Data Preprocessing and Feature Engineering

3.2.1 Data Preprocessing

Firstly, perform data cleaning: delete duplicate records; Exclude records with empty Customer ID (unable to associate with specific individuals); Delete records that are clearly unrelated to purchase predictions or represent abnormal behavior, such as return records with invoice numbers starting with 'C' and entries with negative Quantity. This study focuses on predicting positive purchasing behavior.

Secondly, defining prediction tasks and constructing labels: The core of this study is a binary prediction problem, which predicts whether a customer will make another purchase within a fixed time window in the future (such as the following month). We will set November 1, 2021 to December 9, 2021 as the forecast period, and mark customers who have made at least one purchase during this period as positive samples (y=1), otherwise as negative samples (y=0). All historical data prior to the forecast period (December 1, 2020 to October 31, 2021) were used to construct features for each customer.

Finally, perform dataset partitioning: randomly divide the dataset into a training set and an independent testing set in a 7:3 ratio based on the Customer ID. This partitioning ensures that all data from the same client will only appear in one of the training or testing sets, effectively avoiding the problem of model evaluation results being inflated due to data leakage.

3.2.2 Feature Engineering

Based on the customer's historical transaction records, four categories of 28 numerical features were constructed for each customer:

RFM core features: Recency, frequency, and monetary amount of the most recent purchase.

Behavioral breadth and depth characteristics: the number of unique products purchased, the number of unique product categories visited, and the average number of products purchased per invoice.

Consumption pattern characteristics: average order value, standard deviation of order value, average unit price of goods, standard deviation of unit price, and proportion of discounted goods purchased.

Time pattern characteristics: average purchase interval days, standard deviation of purchase interval days, proportion of purchases made on weekends, distribution of purchases during different time periods such as morning, afternoon, and evening on weekdays.

3.3. Model selection and implementation

Four representative machine learning algorithms were selected for comparative research, and their core formulas were briefly explained:

Logistic regression: serves as a benchmark for linear models. It maps the linear combination of features to probability through the Sigmoid function, as follows. This model is known for its good interpretability (coefficient size and direction representing feature influence) and efficiency.

$$P(y = 1 \mid x) = \frac{1}{1+e-(w^T x+b)} \quad (1)$$

Among them, *x* is the feature vector, *w* and *b* are the model weights and biases.

Support Vector Machine: A powerful classifier based on statistical learning theory. The core idea is to find an optimal hyperplane to maximize the classification interval between samples of different categories. For nonlinear problems, linear separability is achieved by mapping the samples to a high-dimensional space using kernel functions (in this study, radial basis kernels are used). The decision function is:

$$f(x) = \mathrm{sign}(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b) \quad (2)$$

Among them, α_i is the Lagrange multiplier, and K (x_i, x) is the kernel function.

Random Forest: A Representative of Bagging Ensemble Learning. By constructing a large number of decision trees and synthesizing their voting results for prediction. When a single decision tree is growing, the Gini coefficient is often used to select the optimal splitting feature to minimize the "impurity" of nodes.

$$\mathrm{Gini}(D)=1-k=1K(p_k)2 \quad (3)$$

Among them, D is the current node sample set, K is the number of categories, and p_k is the proportion of samples belonging to the k-th category. Random forest effectively reduces model variance and prevents overfitting by introducing dual randomness of samples and features.

XGBoost: An advanced representative of Boosting ensemble learning. It fits the data using an additive model (weighted sum of multiple decision trees) and iteratively adds new trees to fit the residuals from the previous round of predictions. The objective function includes a loss function and a regularization term to control model complexity and prevent overfitting.

$$L^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (4)$$

Among them, l is the loss function (such as logarithmic loss), Ω (ft) is the regularization term used to control the complexity of the t-th tree f. XGBoost optimizes the objective through second-order Taylor expansion approximation, demonstrating excellent performance in both accuracy and efficiency.

All features were standardized before training. Use grid search combined with 5-fold cross validation to optimize key hyperparameters for each model on the training set, such as the regularization coefficient C of logistic regression and SVM, the number and maximum depth of trees in random forests, the learning rate and maximum depth of trees in XGBoost, etc. The final evaluation of model performance is conducted on an independent test set that has never been involved in training and tuning.

4. Result and Discussion

4.1. Result

After completing data preprocessing, feature engineering, and model tuning, the performance of each model was evaluated on a unified test set. The evaluation used five indicators, namely accuracy, precision, recall, F1 score, and AUC (area under the ROC curve), to comprehensively reflect the classification ability of the model. The results are shown in Table 1.

Table 1 Comprehensive Comparison of Prediction Performance of Different Machine Learning Models

Model	Accuracy	Precision	Recall	F1 score	AUC
Logistic regression	0.834	0.701	0.468	0.562	0.792
SVM	0.848	0.732	0.512	0.602	0.821
SF	0.862	0.758	0.570	0.651	0.853
XGBoost	0.876	0.781	0.602	0.680	0.872

Table 1 shows the five core performance indicators of four models on the test set. Accuracy measures the proportion of correct overall classification; Accuracy measures the proportion of customers predicted by the model to actually make a purchase; Recall rate measures the proportion of customers who will actually make a purchase that is successfully predicted by the model; The F1 score is the harmonic mean of precision and recall, used for comprehensive evaluation; The AUC measurement model has the overall ability to rank positive samples higher than negative samples, and the closer it is to 1, the better. As can be seen from the table, XGBoost leads in all indicators.

In addition to prediction accuracy, the computational efficiency of the model is an important consideration in practical applications. Table 2 records the time it takes for each model to complete training and test set prediction in the same hardware environment.

Table 2 Comparison of Model Training and Prediction Efficiency (Unit: Second)

Model	Training time	Prediction time (test set)
Logistic regression	0.8	0.02
SVM	125.3	18.7
SF	15.2	0.35
XGBoost	9.5	0.08

Table 2 illustrates: This table compares the operational efficiency of four models. Training time refers to the total time required to complete model training (including cross validation tuning) on the training set; Prediction time refers to the time it takes to make forward predictions on the test set. Logistic regression has the fastest speed, SVM is the slowest, and Random Forest and XGBoost are in between. Among them, XGBoost achieves the best accuracy while maintaining high training and prediction efficiency.

To understand the decision-making basis of the model, we analyzed the feature importance ranking of each model. Table 3 shows the three most important features identified by logistic regression (based on normalized coefficient absolute values), random forest, and XGBoost.

Table 3 Model Feature Importance Ranking (Top 3)

Ranking	Logistic regression (coefficient absolute value)	SF	XGBoost
1	Recent purchase time (Recency)	Latest purchase time(Recency)	Latest purchase time(Recency)
2	Total consumption amount (Monetary)	Total frequency of purchases	Total frequency of purchases
3	Total frequency of purchases	Total consumption amount (Monetary)	Average order value

Table 3 shows the three most important features that different models consider for prediction. Logistic regression determines importance by the size and direction of feature weights, while Random Forest and XGBoost determine importance by calculating the reduction in impurity (or information gain) caused by features when splitting nodes. Although the model principles are different, "recent purchase time" is unanimously considered as the primary predictor, verifying the core position of R (proximity) in the RFM framework. XGBoost recognizes the importance of the derived feature of "average order value", reflecting its ability to capture more complex patterns.

4.2. Discussion

The recall rate of logistic regression is the lowest (0.468), indicating that its prediction strategy is relatively conservative and tends to misjudge more potential customers as not purchasing. This may be due to its strong assumption of linearly separable data not being valid on complex behavioral data. SVM uses kernel function mapping to find nonlinear interfaces in high-dimensional space, which performs better than logistic regression. However, its performance is highly dependent on the selection of kernel function and penalty parameter C, resulting in high optimization costs. Random forest constructs a large number of decision trees with high diversity through Bootstrap aggregation and random feature subset strategy, and smooths the prediction variance of individual trees through the "voting" mechanism, thus achieving stable and excellent performance. XGBoost uses a Boosting mechanism, where subsequent trees focus on learning difficult samples that were misjudged earlier, and combine the predictions of all trees in the form of additive models. At the same time, the regularization term in its objective function effectively controls complexity, ultimately achieving the best prediction accuracy.

Logistic regression has the fastest training and prediction speed, and is suitable for online reasoning scenarios that require extremely high real-time performance or extremely limited resources. The training time of SVM far exceeds other models, and its time complexity becomes the main bottleneck in large-scale data, with poor scalability. The training of random forests can be highly parallelized and the time is acceptable. XGBoost achieves top-level prediction accuracy while optimizing its weighted quantile sketch, histogram algorithm, and other features, resulting in significantly higher training efficiency than random forests. It demonstrates an excellent balance between accuracy and efficiency, making it highly suitable for large-scale industrial applications.

In terms of model interpretability, the coefficients of logistic regression have the most direct interpretability, and the size and direction of feature weights clearly indicate their marginal impact on purchase probability. The tree model provides global interpretation through feature importance. A highly insightful finding is that regardless of model complexity, 'recent purchase time' remains the most important predictive feature among all models, which strongly validates the classic business rule (R in RFM models) that 'customer recent activity is the best predictor of future behavior' in a data-driven manner. The three core indicators (R, F, M) in the RFM framework consistently rank high in importance ranking, which deeply indicates that domain knowledge based careful feature engineering is the cornerstone of improving the performance of any machine learning model. XGBoost further explores the importance of the derived feature of "average order value", reflecting its ability to capture more subtle consumption patterns.

5. Conclusion

This study systematically compared the performance of four machine learning algorithms, namely logistic regression, support vector machine, random forest, and XGBoost, in predicting consumer purchasing behavior. The experimental results show that under the same feature engineering and evaluation framework, XGBoost algorithm achieves the best balance between prediction accuracy (F1 score 0.680, AUC 0.872) and training efficiency, significantly better than other comparative models, demonstrating the powerful ability of gradient boosting ensemble algorithm in processing complex and non-linear consumer behavior data. Random forests exhibit excellent accuracy and robustness, while logistic regression holds irreplaceable value in specific scenarios due to its outstanding interpretability and computational efficiency. The analysis of feature importance consistently confirms the core predictive power of RFM features such as "recent purchase time", highlighting the importance of combining domain knowledge with data-driven modeling.

Reference

[1] Li L. Analysis of e-commerce customers' shopping behavior based on data mining and machine learning[J]. Soft Computing, 2023: 1-10.

[2] Akram N, Aravindhan K, Sujatha K, et al. Consumer behavior prediction using machine learning algorithms[M]//Exploring psychology, social innovation and advanced applications of machine learning. IGI Global Scientific Publishing, 2025: 109-130.

[3] Lin J. Application of machine learning in predicting consumer behavior and precision marketing[J]. PLoS One, 2025, 20(5): e0321854.

[4] Ebrahimi P, Basirat M, Yousefi A, et al. Social networks marketing and consumer purchase behavior: The combination of SEM and unsupervised machine learning approaches[J]. Big Data and Cognitive Computing, 2022, 6(2): 35.

[5] Mohan L, Devarajan M, Alotoum F J, et al. Advanced Data Analytics for Predicting Consumer Behavior Using Machine Learning and Association Rule Mining[C]//2025 International Conference on Technology Enabled Economic Changes (InTech). IEEE, 2025: 908-914.

[6] Chaubey G, Gavhane P R, Bisen D, et al. Customer purchasing behavior prediction using machine learning classification techniques[J]. Journal of Ambient Intelligence and Humanized Computing, 2023, 14(12): 16133-16157.

[7] Li H. Social Media Data Mining and Online Consumer Behavior Analysis[J]. Procedia Computer Science, 2025, 261: 406-413.

[8] Bhoyar S, Bhoyar P, Shah M A. A machine learning-based predictive approach in evaluating consumer behavior[J]. Journal of Statistics and Management Systems, 2023, 26(8): 1955-1963.

[9] Abdul Aziz M, Mustakim N A, Abdul Rahman S. Decision tree and rule-based classification for predicting online purchase behavior in Malaysia[J]. Malaysian Journal of Computing (MJoC), 2024, 9(2): 1905-1915.

[10] Zvarikova K, Machova V, Nica E. Cognitive artificial intelligence algorithms, movement and behavior tracking tools, and customer identification technology in the metaverse commerce[J]. Review of Contemporary Philosophy, 2022, 21: 171-187.

[11] GhorbanTanhaei H, Boozary P, Sheykhan S, et al. Predictive analytics in customer behavior: Anticipating trends and preferences[J]. Results in Control and Optimization, 2024, 17: 100462.

[12] Xu Z, Zhu G, Metawa N, et al. Machine learning based customer meta-combination brand equity analysis for marketing behavior evaluation[J]. Information Processing & Management, 2022, 59(1): 102800.